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A comparison of three sampling approaches for detecting PRRSV in suckling piglets

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ABSTRACT

Determining whether porcine reproductive and respiratory syndrome virus (PRRSV) is circulating within a breeding herd is a longstanding surveillance challenge. Most commonly, piglets in farrowing rooms are sampled to infer the PRRSV status of the sow herd, with sample size based on the expectation of hypergeometric distribution and piglet selection based on simple random sampling (SRS), i.e., randomly selecting individuals from a population in a manner that all individuals have equal chance of being selected. Conceptually straightforward, the assumptions upon which it is based (homogeneous population and independence of individuals) rarely hold in modern swine facilities. Alternative approaches for sample selection include two-stage stratified sampling (2SS), i.e., randomly selecting litters (first stratum) and randomly selecting piglets (second stratum) within selected litters, and risk-based sampling (RBS), i.e., selecting litters with a higher risk of having viremic piglets, and randomly selecting pigs within those litters. The objectives of this study were to 1) characterize the pattern of distribution of PRRSV-viremic piglets in farrowing rooms and 2) compare the efficiency of SRS, 2SS, and RBS for the detection of PRRSV-viremic piglets. In 12 sow farms, serum samples were collected from all 4510 piglets in 422 litters housed in 23 farrowing rooms and tested for PRRSV RNA. At the population level, the distribution of PRRSV-viremic pigs was analyzed for population homogeneity and spatial clustering. At the litter level, litter size and sow parity were evaluated as risk factors. A non-homogeneous distribution of PRRSV-viremic piglets was observed in nearly all farrowing rooms (15/16), and spatial clustering detected on 11 occasions (11/16). Simulated sampling based on farrowing room data determined that 2SS required 1-to-25 fewer samples than SRS to detect ≥ 1 viremic piglet in 13 of 16 rooms and the same number of samples in 3 rooms. RBS required 1-to-7 fewer samples than 2SS to detect ≥ 1 viremic piglet in 7 of 16 rooms, the same number of samples in 6 rooms, and 1 more sample in 3 rooms. Notably, SRS was less efficient than either 2SS or RBS in detecting PRRSV-viremic piglets in farrowing rooms, regardless of the confidence level. It may be concluded that the core assumptions upon which most current surveillance methods are based do not hold in modern farrowing room facilities. Simulation-based sample size tables for SRS and 2SS are provided.

1. Introduction

Porcine reproductive and respiratory syndrome (PRRS) was first described in 1987 (Keffaber, 1989) with PRRS virus (PRRSV) first isolated in 1991 (Wensvoort et al., 1991). Since then, the virus has been reported in all major swine producing countries, with rare exceptions (Zimmerman et al., 2012). In the United States (U.S.), the Morrison swine health monitoring project (MSHMP) has aggregated PRRS outbreak data from breeding farms and reported weekly summary statistics since 2009 (Tousignant et al., 2015; Arruda et al., 2018a, b; Sanhueza et al., 2019). In 2019, the data provided to MSHMP represented ~3.0 million sows on over 1000 farms, with 24 % reporting PRRS

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outbreaks. Thus, despite concerted prevention and control efforts, PRRS continues to impose economic losses to the U.S. industry estimated at $664 million annually (Holtkamp et al., 2013). For this reason, limiting PRRSV circulation and ultimately eliminating the virus from breeding herds is a common goal.

Breeding herd PRRSV elimination without depopulation is commonly based on a herd closure strategy, with or without whole-herd exposure to live virus (Torremorell et al., 2002). This process involves first overstocking the breeding herd with replacement gilts and then temporarily closing the herd to replacement gilt introduction. The mass exposure of sows to live virus uses the resident PRRSV or a modified-live virus (MLV) vaccine as an attempt to place all animals in the herd on the same timeline for the development of PRRSV herd immunity. Thereafter, consistent production of PRRSV-negative weaned pigs is considered an indication of significant progress towards complete cessation of transmission within the breeding herd. According to the American Association of Swine Veterinarians (AASV) guidelines, achievement of this stage is based on four consecutive negative PRRSV RT-PCR tests of serum samples from 30 due-to-wean piglets at 30-day intervals (Holtkamp et al., 2011). This approach assumes that sampling 30 piglets at one point in time will detect ≥ 1 viremic piglets with 95 % confidence if prevalence is ≥ 10 % (Cannon and Roe, 1982). However, some herds that met these criteria later tested positive for PRRSV, with what appears to be the original outbreak virus, as determined by open reading frame (ORF)-5 sequencing (Linhares et al., 2014), which implies that the virus was present at low prevalence, but undetected in the breeding herd.

Producers and veterinarians need clear and reliable indicators regarding progress towards PRRSV control and elimination, particularly during the final stages of elimination. To meet this need, the objectives of this study were to characterize the distribution of PRRSV-viremic piglets in litters within farrowing rooms and, based on the findings, model and compare the efficiency of PRRSV farrowing room sampling strategies based on individual piglet serum sampling.

2. Material and methods

2.1. Study design

A cross-sectional study was performed in 12 breed-to-wean sow farms (Table 1) in which serum samples (n = 4510) were collected from all piglets in selected litters (n = 422) in 23 farrowing rooms and tested individually for PRRSV RNA (Table 2). Participants in the study represented a convenience sample of farms located in Iowa, Illinois, and Minnesota USA. The management responsible for each farm agreed to participate and provide the required data. Data collected included sow parity (provided for 269 of 422 litters), piglet age at testing, number of piglets within a litter, and farrowing crate location within the farrowing room. The test results were modeled to estimate the number of litters and piglets within litters needed to detect ≥ 1 PRRSV-positive piglets in a farrowing room with 90 %, 95 %, and 99 % confidence, using different sampling approaches. This study was approved by the Iowa State University Office for Responsible Research under protocol #4−16-8240-S.

Table 1
Specific characteristics of PRRSV-endemic (as defined by the herd veterinarian) sow herds participating in the study.

Farm	Sow inventory (n)	Farrowing schedule	Piglet age at sampling	Immunologic intervention
A	1100	Continuous	~ 21 days	PRRSV killed vaccine[1]
B	6500	Continuous	~ 21 days	LVI[2]
C	350	2-week batch farrow	~ 21 days	MLV[4]
D	3000	Continuous	~ 21 days	PRRSV MLV[3] quarterly in reproductive females
E	1200	2-week batch farrow	~ 21 days	Sows: LVI[2] Gilts: PRRSV killed vaccine[1] + LVI[2]
F	3600	Continuous	~ 21 days	LVI[2]
G	1100	Continuous	~ 21 days	LVI[2]
H	2200	Continuous	~ 21 days	LVI[2]
I	5200	Continuous	~ 3 days	LVI[2]
J	2800	Continuous	~ 3 days	LVI[2]
K	1750	Continuous	~ 3 days	LVI[2]
L	6000	Continuous	~ 3 days	–

[1] MJPRRS Vaccine, Phibro Animal Health Corporation (Teaneck, NJ, USA).
[2] LVI - live PRRS virus inoculation.
[3] Fostera PRRS, Zoetis Animal Health (Parsippany-Troy Hills, NJ, USA).
[4] Ingelvac PRRS® MLV (St. Joseph, MO, USA).

2.2. Missing data

In most rooms, some crates were not sampled due to being empty or piglets not sampled. Those are as follows - Farm A room 1, two crates not sampled, one crate empty; farm A room 2: 4 crates not sampled, three crates empty; Farm A room 3: one crate not sampled; Farm A room 4: thirteen crates not sampled; Farm B: sixteen crates not sampled; Farm C room 1: three crates empty, Farm C room 2: two crates empty; Farm D: two crates empty; Farm E: seven crates not sampled; Farm F: one crate empty; Farm H: three crates not sampled, two crates empty; Farm I: fourteen crates not sampled, two crates empty; Farm J room 1: four crates not sampled; Farm J room 2: two crates not sampled, one crate empty; Farm K room 1: thirteen crates not sampled; Farm K room 2: one crate not sampled; Farm K room 3: sixteen crates not sampled; Farm k room 4: one crate not sampled; Farm L room 1: one crate not sampled; Farm L room 2: seventeen crates not sampled.

2.3. Sample collection and handling

All samples were collected between May 2017 and September 2019. Blood samples were collected from piglets via jugular venipuncture using a single-use sterile blood collection system (B.D. Vacutainer™, Thermo Fisher Scientific™, Waltham, MA, USA). Immediately following collection, samples were labeled by farm, collection date, and farrowing crate information (position within a farrowing room). Collected samples were maintained at 4 °C–8 °C and transported to a National Animal Health Laboratory Network-accredited diagnostic laboratory for processing within 24 h. In the laboratory, samples were centrifuged for 10 min at 1600 x *g*, and then serum samples were submitted for individual PRRSV RNA testing.

2.4. Diagnostic testing

Serum samples were tested for PRRSV RNA by real-time reverse-transcriptase polymerase chain reaction (RT-rtPCR) using routine testing procedures at the Iowa State University (Ames, IA) or University of Minnesota (St. Paul, MN) Veterinary Diagnostic Laboratories. In brief, nucleic acids were extracted using the KingFisher® Flex automated magnetic particle processor system (ThermoFisher Scientific™) and a commercial extraction kit MagMAX™ Pathogen RNA/DNA Kit (ThermoFisher Scientific™), as directed by the manufacturer. PRRSV rRT-PCR was performed on nucleic acid extracts using the MagMAX™ North American (N.A.) and European (E.U.) PRRSV-specific PCR assay (ThermoFisher Scientific™).

2.5. Statistical analysis

The data were analyzed in three steps. First, the distribution of PRRSV-viremic piglets in farrowing rooms was evaluated for population homogeneity and spatial clustering. By default, only the 16 rooms housing viremic piglets were included in this analysis (Farm A rooms 1, 2, 3, and 4; Farm C rooms 1 and 3; Farm E; Farm G room 2; Farm H; Farm I; Farm J rooms 1 and 2; Farm K rooms 1, 2, 3, and 4 – Table 2). Second,

Table 2
PRRSV-viremic piglets in breeding herds: count of positive litters and positive piglets within litters.

Farm	Room	Count of PRRSV-positive litters by number of positive piglets within positive litters														Positive litters	Total litters	Positive piglets	Total piglets
		1	2	3	4	5	6	7	8	9	10	11	12	13	14				
A	1	1	–	1	3	–	2	2	2	–	1	2	–	–	–	14	17	90	157
	2	2	–	–	1	–	–	1	–	–	–	–	–	–	–	4	20	13	201
	3	2	–	–	–	–	–	–	1	1	1	–	–	–	–	5	17	29	153
	4	–	1	–	–	–	–	–	–	–	–	–	–	–	–	1	5	2	53
B	1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	40	0	450
C	1	–	–	–	–	–	–	–	1	–	–	–	–	–	–	1	7	8	60
	2	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	7	0	78
	3	–	–	–	1	–	–	–	–	–	–	–	–	–	–	1	10	4	117
D	1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	26	0	315
E	1	3	1	–	–	1	–	–	1	–	2	–	–	–	–	8	13	38	135
F	1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	55	0	681
G	1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	22	0	215
	2	1	1	–	–	–	–	–	–	–	–	–	–	–	–	2	22	3	240
H	1	2	3	–	1	1	–	–	–	–	–	1	–	–	–	8	19	30	176
I	1	2	–	1	3	3	–	2	–	1	–	1	–	–	–	13	24	66	271
J	1	2	3	–	1	–	2	1	2	–	1	2	2	–	1	17	20	117	211
	2	5	1	1	3	2	2	2	–	–	–	–	–	–	–	16	21	58	163
K	1	2	–	1	–	–	–	–	–	1	–	–	–	–	–	4	7	14	67
	2	–	1	1	–	1	–	–	–	–	–	–	–	–	–	3	19	10	222
	3	–	–	–	1	1	–	–	–	–	–	–	–	–	–	2	4	7	39
	4	8	–	–	1	1	1	1	–	–	1	–	–	–	–	13	17	36	173
L	1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	23	0	256
	2	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0	7	0	77
Total litters		30	11	7	14	9	8	8	7	3	6	6	2	0	1	112	422	525	4510

the data were analyzed to identify risk-factors associated with viremic piglets. For this analysis, only litters for which litter size and sow parity information was available were included (269 litters from farms A, B, C, D, F, I, and J). Lastly, three sampling strategies for detecting PRRSV viremic pigs in farrowing rooms were compared. Statistical analyses of the first and the third steps were conducted using R version 3.5.2. (R Core Team, 2018). The risk factor analysis in step 2 was conducted using SAS 9.4 (SAS Institute Inc., Cary, NC, USA). $P \leq 0.05$ was considered as statistical significance.

2.5.1. Population homogeneity and spatial clustering

The distribution of PRRSV-viremic piglets among litters within farrowing rooms was evaluated for population homogeneity and for spatial clustering (Farm A rooms 1, 2, 3, and 4; Farm C rooms 1 and 3; Farm E; Farm G room 2; Farm H; Farm I; Farm J rooms 1 and 2; Farm K rooms 1, 2, 3, and 4 – Table 2). Population homogeneity implies that within-litter prevalence should be similar within each farrowing room. Spatial clustering occurs when viremic piglets are in closer proximity than would be expected by chance.

To test for population homogeneity, N was the total number of piglets within a room and M the total number of litters. Homogeneous disease distribution implies that the viremic status of each piglet in the room (either being viremic or not) follows an identical probabilistic distribution, with the total number of positive piglets determined by prevalence. Under the hypothesis of homogeneity, the probability of disease occurrence of the j -th piglet ($j = 1, 2, \ldots, N$) is the same across all piglets in the room; this implies that the within-litter disease prevalence should likewise be similar. Hence, with p_i representing the within-litter prevalence of the i -th litter, the null and alternative hypotheses were:

$$H_0 : p_1 = p_2 = \ldots = p_M \quad (1)$$

H_a : there exists at least one inequality in H_0.

In this scenario, the null hypothesis was rejected if the p-value of Fisher's exact test is ≤ 0.05 (i.e., disease occurrence is not homogeneous); otherwise, the null hypothesis is accepted.

The presence of spatial clustering was assessed using a permutation test to compare the locations of observed viremic piglets to locations determined by random allocation (Turnbull et al., 1990). As illustrated in Fig. 1, piglet location was based on the two-dimensional x–y coordinates of its farrowing crate, assuming the crates were aligned in unit grid (Sun et al., 2019). The spatial center of disease for each farrowing room, or disease centroid, was calculated as the arithmetic mean of the x and y coordinates of all viremic piglets. All piglets within a farrowing crate shared the same coordinates. The following permutation test was performed to test for the null hypothesis of random disease distribution versus the alternative hypothesis of clustered disease distribution:

1 For each observed farrowing room, the average Euclidean distance ($\overline{D}_{obs}$) between the viremic piglets and the disease centroid was calculated.
2 To test for clustering, all piglets in the farrowing room were randomly permuted (re-allocated) to litters via computer simulation while keeping the litter sizes equal to the observed data. For the b -th simulated data, the metric in step 1 ($\overline{D}_b^*$) was calculated.
3 Step 2 was repeated for B = 10,000 iterations. The p-value was the proportion of $\overline{D}^*$ that were as extreme as, or more extreme than, $\overline{D}_{obs}$: $p - \text{value} = \frac{1}{B}\sum_{b=1}^{B}(\overline{D}_b^* < \overline{D}_{obs})$. The distribution of viremic piglets was considered clustered if $p < 0.05$.

2.5.2. Risk-factor analysis

Parity and litter size (at sampling time) information was available for 269 litters containing 2910 piglets from farms A rooms 1, 2, 3, and 4, B, C rooms 1, 2, and 3, D, F, I, and J rooms 1 and 2, totaling 13 rooms.

Risk factor analysis of the effect of litter size, parity, and their interaction on the probability that a litter would contain ≥ 1 PRRSV-viremic piglets was performed using mixed-effect logistic regression analysis with room and farm considered random effects (PROC GLIMMIX). Besides, a generalized linear model was used to evaluate the effect of litter size, parity, and their interaction on the PRRSV within-litter prevalence with room and farm considered random effects (PROC GLIMMIX). Unlike logit transformation in the logistic regression, within-litter prevalence was log-transformed for ease of result interpretation. Model diagnostics were conducted with residual plots in SAS.

In both analyses, the fixed effects were binary: first parity sow (Y/N) and litter size (small litter size: ≤ 12 piglets; large litter size: > 12

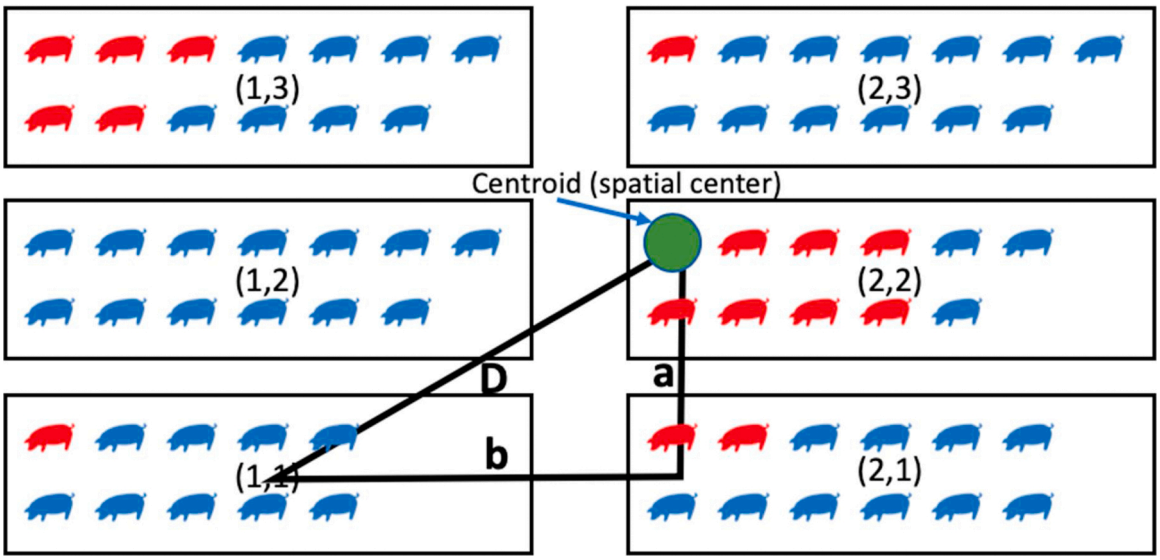

Fig. 1. Elements used in the cluster analysis.
In the example, the disease centroid (spatial center) is located at x = 1.6, y = 2.2. The Euclidian distance (D) between the disease centroid and the viremic piglet in the litter located at x = 1, y = 1 is calculated as ($D = \sqrt{1.2^2 + 0.6^2}$).

piglets). The litter size threshold of ≤ 12 piglets was established on the basis of the contrast of within-litter prevalence by litter size, as shown in the histogram (Fig. 2), i.e., few positive litters were observed in litters of ≥ 12 piglets.

2.5.3. Comparison of sampling strategies

The power of detection (POD) for simple random sampling (SRS), two-stage stratified sampling (2SS), and risk-based sampling (RBS) was estimated by computer simulations based on the study dataset, i.e., 525 viremic piglets among 2442 piglets in 296 litters in 16 rooms in 8 breed-to-wean sow farms. For SRS, a given number of piglets (*n.sample*) were randomly selected without replacement within a room. If ≥ 1 positive piglet(s) was detected in one simulation, the trial was regarded as a "success." For each *n.sample*, the proportion of successful trials (10,000 iterations) was the POD. This process was continued until the *n.sample* that achieved 90 %, 95 %, and 99 % POD was determined. The POD for 2SS and RBS differed only in the simulated sampling procedure. For 2SS, the first stage was to randomly select litters (strata), and the second stage was to randomly select piglets within litters. If *n.sample was* less than or equal to the total number of litters, then *n.sample* litters were selected without replacement in the first stage, and one piglet was randomly selected within each litter in the second stage. If *n.sample* was greater than the total number of litters, then all litters were included in the first stage, and a single piglet was randomly sampled from each litter in the second stage. Thereafter, the remaining samples were selected randomly (litter and piglet within litter) until *n.sample* piglets were sampled (without replacement). As an example consider a room with 6 litters and 10 piglets in each litter. Suppose *n.sample* = 5, which is less

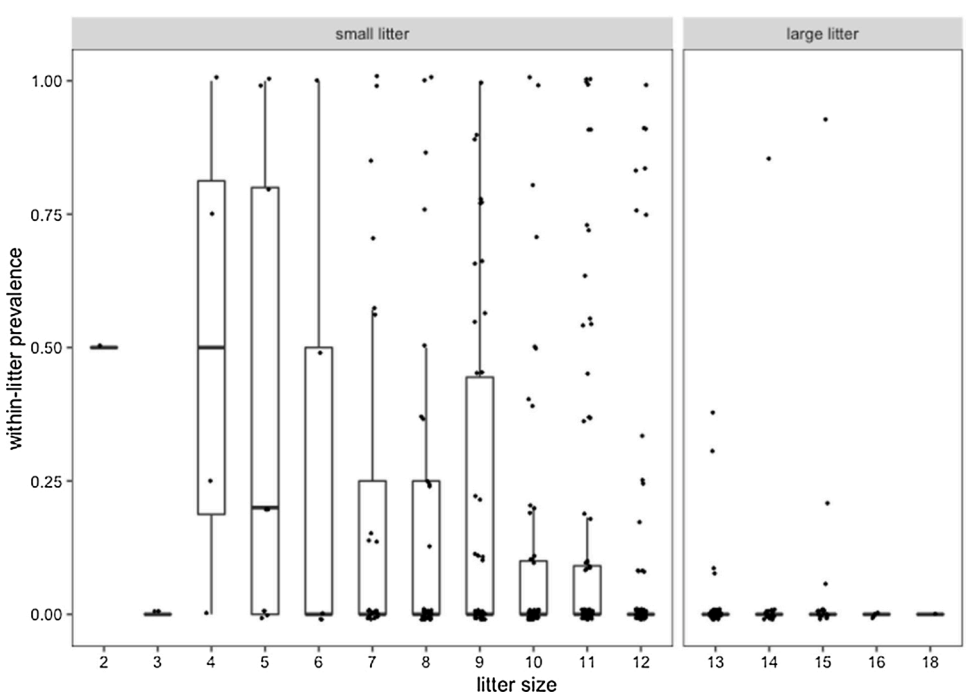

Fig. 2. Within-litter PRRSV prevalence by litter size.

Table 3
Analysis of population homogeneity and clustering of PRRSV-viremic piglets by farm and room.

Farm	Room	Positive piglets	Population homogeneity analysis				p-value Fisher's exact test	Clustering analysis	
			Expected positive litters[1]		Observed positive litters				
			n	Avg. No. of positive piglets per litter	n	Avg. No. of positive piglets in positive litters		Clustered (Y/N)	p (probability)
A	1	90	17	0.57	14	0.66	< 0.01	N	0.684
	2	13	20	0.06	4	0.36	< 0.01	Y	0
	3	29	17	0.19	5	0.58	< 0.01	Y	0
	4	2	5	0.04	1	0.20	> 0.05	N	0.185
C	1	8	7	0.13	1	1.00	< 0.01	Y	0
	3	4	10	0.03	1	0.33	< 0.01	Y	0.001
E		38	13	0.28	8	0.44	< 0.01	Y	0
G	2	3	22	0.01	2	0.14	< 0.05	N	0.219
H		30	19	0.17	8	0.38	< 0.01	Y	0.001
I		66	24	0.24	13	0.49	< 0.01	Y	0
J	1	117	20	0.55	17	0.65	< 0.01	Y	0
	2	58	21	0.36	16	0.46	< 0.01	Y	0.03
K	1	14	7	0.21	4	0.42	< 0.01	Y	0
	2	10	19	0.05	3	0.37	< 0.01	N	0.315
	3	7	4	0.18	2	0.64	< 0.01	Y	0.024
	4	36	17	0.21	13	0.28	< 0.01	N	0.329

[1] Expected based on the assumption of population homogeneity.

than the total number of litters (6). Then, the first stage of the two-stage sampling approach is to sample 5 litters randomly from the 6 litters. For the selected 5 litters, we sample 1 piglet, to achieve the goal of *n.sample* = 5. Now, if *n.sample* = 8, which is more than the total number of litters (6). Then, the first stage is to include all 6 litters, and randomly sample 1 piglet from each litter, to get 6 samples. For the remaining 2 piglets, just like condition 1, we can randomly sample 2 litters out of 6, and sample 1 piglet from the selected 2 litters. Likewise, RBS used a two-stage approach, but unlike the random selection in the first stage of 2SS, RBS first targeted high-risk litters (as determined by the results of the risk factor analysis, i.e., parity and/or litter size at sampling). If no more high-risk litters were available, 2SS was then used to complete the number of piglets to be sampled (*n.sample*).

To expand sample size estimates beyond the study data, simulations were performed in *R* to estimate sample sizes for the detection of ≥ 1 positive piglet over a range of PRRSV prevalence (1%–25%), a range of piglet population sizes (100–600), and power of detection (90 %, 95 %, 99 %) for SRS and 2SS sampling methods. RBS was not included because considering the various risk factor combinations when simulating data could be over complicated. Simulations were performed for two scenarios, i.e., random distribution and spatial clustering of viremic piglets in the fewest possible litters. For each of 1000 iterations, viremic piglets were "re-distributed" (randomly or clustered), after which 1,000″samplings" were performed. Within the simulation modeling, each litter was assumed to contain 10 piglets.

3. Results

In 12 PRRSV-endemic breeding farms (A through L, Table 2), positive litters (n = 112, 26.5 %) contained a total of 1080 piglets, of which 525 were viremic (48.6 %). Among positive litters, 26.8 % (n = 30) had one PRRSV positive piglet; 28.6 % (n = 32) had two to four positive piglets; and 44.6 % (n = 50) had five to 14 positive piglets (Table 2). Statistical analysis of the distribution of PRRSV-viremic piglets within rooms at each farm revealed a non-homogeneous distribution in all rooms, with the exception of Farm A room 4 ($p > 0.05$) (Table 3). Cluster analysis revealed spatial clustering of viremic piglets in 11 of 16 rooms ($p \leq 0.05$) (Table 3).

Visualizations of the distribution of within litter prevalence for all data were shown in Fig. 2 (by litter size) and Fig. 3 (by parity). Without considering the litters with missing parity information, summary statistics are as follows: there were 15 positive litters among 45 (33.3 %) first parity sows versus 57 positive litters out of 224 (25.4 %) multiparous sows. First parity sows had a higher proportion of viremic piglets in their litters (17.4 %) than multiparous sows (14.8 %). Likewise, there were 67 positive litters among 202 (33.2 %) small litters (≤ 12 piglets, average 9.83 piglets) versus 5 positive litters from 67 (7.5 %) large litters (> 12 piglets, 13.80 piglets). Within-litter prevalence in small litters was 19.0 % versus 3.8 % in large litters.

The generalized linear models provided results at a higher resolution by taking the litter size (small / large), parity (first parity / multiparous), and their interaction as fixed effects; room and farm as random effects. Based on the mixed-effect logistic regression model analysis on the data excluding missing parity records (269 litters with 2910 piglets), the odds of containing viremic piglets in small litters was 4.03 (95 % CI: 0.84–19.23, p = 0.08) times than in large litters with moderate significance. However, for small litters only, the odds of first parity sows having litters with > = 1 viremic piglet was 4.32 (95 % CI: 1.36–13.71, p = 0.01) times than multiparous sows. According to the model of the prevalence, within-litter prevalence in small litters was 4.42 (95 % CI: 1.65–11.81, p = 0.003) times than large litters on average. For small litters only, within-litter prevalence of first parity sows was 1.76 (95 % CI: 1.41–2.20, $p < 0.0001$) times than large litters.

Based on the data observed in this study, Table 4 shows the sample size necessary to detect ≥ 1 viremic piglet as a function of sampling strategy based on the observed farm and room data. Based on simulation modeling, sample size estimates for the detection of ≥ 1 viremic piglet by PRRSV prevalence, population size, and power of detection (90 %, 95 %, 99 %) are given in Tables 5–7 for SRS and 2SS sampling methods.

4. Discussion

The expansion in breeding herd inventory that marked the 1990s (USDA, 2017) led to the construction of farrowing facilities with multiple rooms, each with many farrowing pens/crates. The effect of these changes on pathogen transmission and sampling for pathogen detection have largely been understudied, yet needs to be understood and taken into account to achieve effective disease surveillance. An efficient sampling approach for detecting PRRSV-viremic piglets in presumably PRRSV-negative herds or herds going through the process of PRRSV control or elimination is a longstanding need for swine producers and veterinarians. The challenge lies in finding the right balance between the investment of surveillance and the risk of failing to detect PRRSV-circulation. The "traditional" approach to farrowing room surveillance has based sample size on the hypergeometric distribution (probability of success in random draws, without replacement, from a

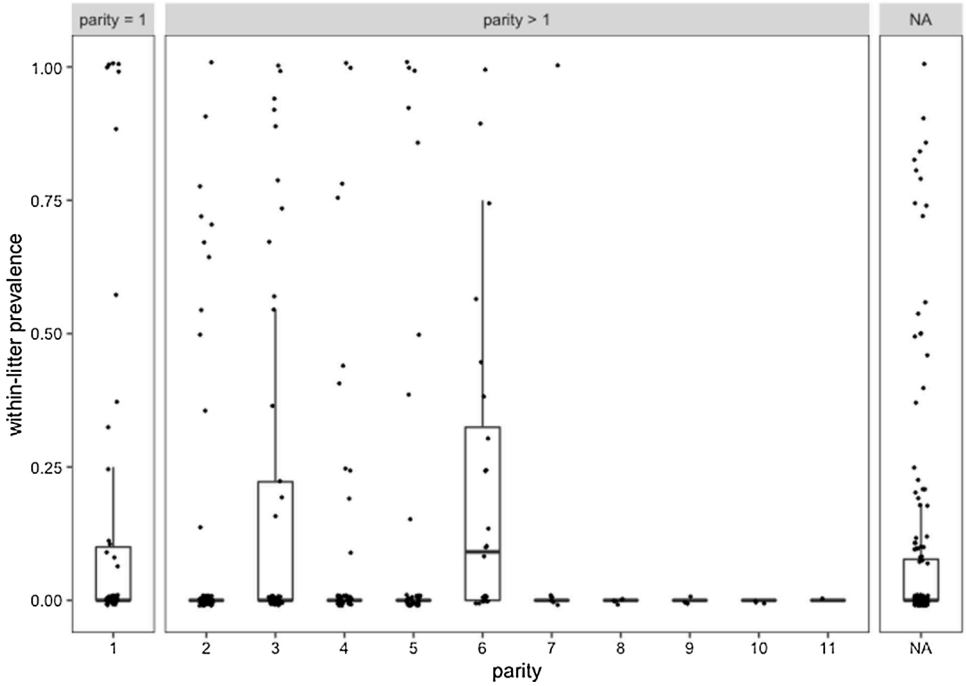

Fig. 3. Within-litter prevalence by sow parity.

Table 4
Sample size to detect ≥ 1 viremic piglet using simple random sampling (SRS), two-stage sampling (2SS), or risk-based sampling (RBS) based on observed data.

PRRSV viremic piglets (%)			57.3	55.5	35.6	28.1	24.4	20.9	20.1	17.9	17.9	17	13.3	6.5	4.5	3.8	3.4	1.3
Farm-room			A-1	J-1	J-2	E	I	K-1	K-4	K-3	A-3	H	C-1	A-2	K-2	A-4	C-3	G-2
Litters, n			17	20	21	13	24	7	17	4	17	19	7	20	19	5	10	22
Population Size			157	211	163	135	271	67	179	39	151	176	60	201	222	53	117	240
Confidence level	90 %	SRS	3	3	6	8	9	10	10	10	11	13	14	32	44	37	50	127
		2SS	3	3	6	7	6	6	8	4	10	11	7	14	30	33	47	97
		RBS	3	3	4	ND*	1	ND	ND	ND	11	ND	7	15	ND	34	44	ND
	95 %	SRS	4	4	7	9	11	12	12	13	13	17	19	41	55	42	58	139
		2SS	4	4	7	8	8	8	10	5	11	13	7	17	37	38	56	114
		RBS	3	4	5	ND	1	ND	ND	ND	12	ND	7	17	ND	37	53	ND
	99 %	SRS	6	6	10	14	14	16	17	18	19	23	26	57	76	48	79	180
		2SS	5	6	11	11	11	11	16	7	15	16	7	19	52	44	67	141
		RBS	4	6	7	ND	1	ND	ND	ND	15	ND	7	20	ND	43	64	ND

* ND = Analysis Not Done due to lack of parity information.

finite population with a specific feature) and piglet selection on simple random sampling (subset chosen randomly from a larger population in a manner that each individual has an equal chance of being selected) - an approach that assumes a homogeneous distribution of PRRSV within the population.

The first objective of this project was to understand the pattern of PRRSV-viremic piglets' distribution within in modern US farrowing rooms using homogeneity and clustering analyses. Homogeneity in PRRSV distribution would imply that each piglet in the farrowing room had an identical and independent chance of being viremic.

The idea of heterogeneity in disease distribution in groups of due-to-wean piglets is not new (Fano et al., 2007; Pieters et al., 2014); however, the heterogeneous distribution of PRRSV among litters in the field had not been previously documented. Likewise, PRRSV clustering is not a new concept, but it is most commonly described for populations in larger geographic areas, such as farms within regions (Arruda et al., 2016, 2017; Arruda et al., 2018a, b). More recently, clustering within wean-to-finish barns was described as a spatiotemporal phenomenon (Rotolo et al., 2017).

The current American Association of Swine Veterinarians requirement for achieving "PRRSV stable" status relies on the absence of viremic piglets in four consecutive monthly samplings of 30 piglets each (Holtkamp et al., 2011). Underlying this approach is the assumption that low prevalence is an indication that PRRSV is being eliminated from the population. However, in one study, the original outbreak virus (based on ORF5 sequencing) was recovered from PRRSV stable herds; thus, low prevalence may not always lead to elimination (Linhares et al., 2014). Further, a growing body of information suggests that PRRSV can be present in cohorts of piglets at a prevalence < 10 %. For example, Cano et al. (2008) sampled all piglets in randomly selected litters following whole-herd inoculation with live PRRSV. At 4 and 12 weeks, respectively, prevalence decreased from 8% to 2% in liveborn piglets and 23 % to 7% in due-to-wean piglets. Graham et al. (2013) tested serum samples from all piglets in 44 litters (n = 454) in a 1100-sow breed-to-wean farm 60 days after all sows were inoculated with live PRRSV and detected 3 (0.7 %) viremic piglets by PRRSV RT-PCR. Vilalta et al. (2018) tested piglets over time post-outbreak in a 6000-sow breed-to-wean farm, reporting that prevalence between weeks 11 and 23 ranged from 0.9 % to 6.5 %. Collectively, these studies demonstrated that it is not uncommon for PRRSV infection to exist in suckling piglet populations at < 10

Table 5
Sample size requirements for detection of PRRSV-viremic piglets at selected prevalence scenarios with 2 different distribution of viremic piglets at 90 % power of detection.

			Prevalence (%)								
Population size*	Sampling strategy	Distribution of viremic piglets	25	20	15	10	5	4	3	2	1
100	SRS	Clustered	8	11	13	19	37	43	53	70	90
		Random	8	10	14	19	39	43	54	66	91
	2SS	Clustered	6	7	9	9	29	37	49	63	92
		Random	7	11	12	18	35	37	49	70	88
200	SRS	Clustered	8	11	14	20	38	45	61	89	135
		Random	9	10	14	21	40	50	65	86	132
	2SS	Clustered	7	9	11	14	18	33	46	75	131
		Random	9	10	15	18	36	48	60	82	130
400	SRS	Clustered	9	12	15	22	42	54	69	98	179
		Random	8	11	15	23	43	53	67	98	172
	2SS	Clustered	7	9	12	17	28	33	36	62	151
		Random	8	11	14	22	40	49	66	93	168
600	SRS	Clustered	9	11	15	22	42	53	69	106	193
		Random	9	11	15	23	42	50	70	101	186
	2SS	Clustered	8	10	12	19	32	38	46	53	138
		Random	8	9	14	21	34	45	74	89	167

* Number of piglets in the population being sampled.

Table 6
Sample size requirements for detection of PRRSV-viremic piglets at selected prevalence scenarios with 2 different distribution of viremic piglets at 95 % power of detection.

			Prevalence (%)								
Population size*	Sampling strategy	Distribution of viremic piglets	25	20	15	10	5	4	3	2	1
100	SRS	Clustered	11	13	17	23	44	52	61	79	96
		Random	10	12	16	26	43	52	63	78	96
	2SS	Clustered	7	8	10	10	36	45	57	72	96
		Random	9	12	14	22	37	45	51	74	96
200	SRS	Clustered	11	13	19	26	49	60	78	103	154
		Random	10	13	18	26	50	63	77	104	153
	2SS	Clustered	9	10	12	15	20	37	58	83	147
		Random	10	11	17	24	41	60	72	92	130
400	SRS	Clustered	12	14	19	28	54	68	85	119	210
		Random	11	15	20	27	53	67	85	122	210
	2SS	Clustered	9	11	15	21	31	36	38	72	167
		Random	11	14	17	24	45	60	75	97	177
600	SRS	Clustered	11	15	19	29	55	70	88	129	236
		Random	11	14	19	28	53	69	87	128	235
	2SS	Clustered	10	12	16	23	37	45	50	56	171
		Random	10	13	16	24	45	58	76	109	182

* Number of piglets in the population being sampled.

Table 7
Sample size requirements for detection of PRRSV-viremic piglets at selected prevalence scenarios with 2 different distribution of viremic piglets at 99 % power of detection.

			Prevalence (%)								
Population size*	Sampling strategy	Distribution of viremic piglets	25	20	15	10	5	4	3	2	1
100	SRS	Clustered	16	18	28	33	60	64	76	88	99
		Random	16	18	24	34	60	67	78	91	99
	2SS	Clustered	9	9	10	10	46	58	69	85	99
		Random	12	14	22	34	46	51	69	85	97
200	SRS	Clustered	15	19	26	38	68	78	103	137	180
		Random	16	20	26	39	70	85	104	128	182
	2SS	Clustered	11	13	15	18	20	51	75	110	173
		Random	15	18	25	34	60	80	95	104	169
400	SRS	Clustered	16	22	27	41	69	89	118	172	273
		Random	17	22	29	39	75	90	122	170	265
	2SS	Clustered	13	15	20	26	36	40	40	93	205
		Random	14	20	22	34	60	78	109	150	244
600	SRS	Clustered	16	21	27	41	80	97	121	191	299
		Random	18	21	28	44	80	100	128	182	317
	2SS	Clustered	14	17	21	28	47	53	58	60	230
		Random	14	16	23	37	59	78	113	161	250

* Number of piglets in the population being sampled.

% prevalence, i.e., levels below the level of detection at the recommended sampling levels.

Cumulatively, heterogeneity and clustering of viremic piglets in farrowing rooms explain the challenges inherent in PRRSV detection, particularly as the disease prevalence declines during elimination programs. Therefore, the second objective of this project was to compare the detection efficiency of three sampling strategies and identify the method that required the fewest samples to detect ≥ 1 viremic piglets.

All probability-based sampling techniques are derived from simple random sampling (Salman, 2003). Simple random sampling assumes that populations are homogeneous, i.e., samples are selected such that each sampling unit in the population has an equal opportunity of being chosen (Cannon and Roe, 1982). This requires a sampling frame and a method of randomization (Salman, 2003). The advantages of simple random sampling include conceptual simplicity, ease of use in small populations, and straightforward sample size calculations for a defined prevalence and probability of detection. A major impediment to the correct implementation of simple random sampling on the farm is the requirement for a sampling frame (a list of all animals in the population from which a sample can be taken), a prerequisite that becomes increasingly cumbersome as the population increases in size.

Naturally occurring homogeneous populations are rare (Finkelstein and Cha, 2013); more typically, subpopulations exist within the larger population. Under these circumstances, estimating population parameters or detecting disease using simple random sampling is problematic because of over- or under-sampling in subgroups, particularly when sample size is small relative to the total population (Cannon and Roe, 1982). Stratified sampling addresses population heterogeneity by dividing sampling units into strata, i.e., non-overlapping homogeneous subgroups (Cochran, 1977; Mazzocchi, 2008; Chambers and Clark, 2012). Sampling independently from each stratum, or from a randomly selected subset of strata, controls for heterogeneity, improves representativity and produces more precise population estimates (Mazzocchi, 2008). Hence, in a farrowing room, two-stage stratified sampling is based on randomly selecting litters and then piglets within litters.

Risk-based sampling targets members of the population considered to be more likely to possess the target of interest, thereby increasing surveillance efficiency (fewer samples with higher probability of detection). In this study, parity and litter size (at sampling time) were identified as significant risk factors for having litters with PRRSV-viremic piglets and used to guide sample selection in a two-stage process. Research on litter size as a risk factor for pathogen detection is sparse, but PRRSV infection has been associated with smaller litters, at birth or weaning, partly due either to intrauterine infections or increased pre-weaning mortality, respectively (Dee and Joo, 1994). Likewise, parity one females and their piglets are reportedly at a higher risk of infection and clinical disease from porcine parvovirus (Truyen and Streck, 2019), porcine hemagglutinating encephalomyelitis virus (Sasseville et al., 2001; Alsop, 2006; Quiroga et al., 2008), and *Mycoplasma hyopneumoniae* (Calsamiglia and Pijoan, 2000). In general, risk-based sampling works best when the overall prevalence of the infection is low (Salman, 2003), as expected during late stages of a PRRSV elimination program. Disadvantages of the approach include the requirement to continually "update" the process with current risk factor assessments and the fact that individuals or litters not considered to be at risk, and therefore not be included in the sampling, can nevertheless be positive for the target of interest.

The simulation model-based sample sizes reported in Tables 5–7 were consistent with the estimates derived from the field data (Table 4). Because the simulation modeling was able to encompass a broader range of population sizes and prevalence, these Tables will provide sample size guidelines for veterinarians and producers interested in surveilling piglet populations. In the majority of scenarios, 2SS was more efficient than SRS, i.e., required fewer samples, to detect ≥ 1 PRRSV viremic piglet regardless of prevalence, population size, or power of detection.

5. Conclusions and implications

This study established that PRRSV-viremic piglets in farrowing rooms are not randomly distributed among litters; rather, they consist of a heterogeneous population that is often clustered. This presents a distinct challenge for routine surveillance methods intended to detect PRRSV-infected piglets. Therefore, three surveillance sampling approaches were compared in terms of surveillance efficiency. A direct comparison of the three approaches showed that simple random sampling was the least efficient strategy for the detection of PRRSV-viremic piglets. Two-stage and risk-based sampling were nearly equivalent, with a slight advantage in favor of detection by risk-based sample. Thus, sampling piglets in high-risk litters, if present, may enhance detection when using a two-stage stratified sampling strategy. Ultimately, surveillance sample size and piglet selection decisions must be tailored to individual circumstances. To that end, the information presented in this study, e.g., the sample size guidelines provided in the tables, will aid producers and veterinarians in developing efficient and effective survey protocols.

Declaration of Competing Interest

The authors report no declarations of interest.

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